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Diagnosing a Register-Pragmatic Blind Spot in Javanese Hate Speech Detection via LLM-Generated Register-Stratified Stimuli

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Abstract

Javanese, spoken by roughly 84 million people, poses a distinct hate-speech detection challenge: its honorific register system (*unggah-ungguh*) lets identical hostile content surface as vulgar *ngoko* or as deceptively refined *krama*. *Krama* hostility is diglossically confined to formal, face-to-face contexts and thus absent from social-media corpora, leaving corpus-based detection blind to an entire class of Javanese hate. This study aims to diagnose that blind spot and build a reproducible instrument for measuring it without naturally occurring data. We construct a register-stratified stimulus set via large language models across four register-pragmatic niches and nine target groups (a 36-cell matrix), validate authenticity with three native and near-native raters, and probe five detector models. The strongest generator, DeepSeek, is judged 97–100% authentic by all three raters; pooled authenticity is rater-dependent (45–91%). The local model Qwen3-14B fails entirely at *krama*, producing Indonesian. Detection reveals a near-universal blind spot: only 2 of 45 verdicts (4%) flag ironic *krama* stimuli as hate, versus 100% for *ngoko*-direct hate; human raters converge on the same uncertainty. Politeness alone does not blind detectors — *krama-report* without irony is caught at 78–89%; the failure is pragmatic implicature. Two independent measurements corroborate this scarcity: zero consensus *krama*-hate in 735 labeled texts, and 0.093% Javanese presence in 1.42 million tweets. The resulting taxonomy, construction method, and detection probe contribute a reusable evaluation methodology for informatics research on NLP content-moderation systems, extensible to other diglossic, register-rich languages.

Keywords : detection blind spot, diglossia, hate speech detection, Javanese, large language model, register

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1. INTRODUCTION

Javanese is spoken by roughly 84 million people and carries significant online hate speech directed at ethnic minorities (Madurese, Chinese-Indonesian, Arab communities), religious groups, women, and LGBTQ+ individuals [1], [2], [3]. Yet the language is severely under-resourced in NLP relative to general-Indonesian resources [4], [5], [6], [7], [8], and the few published hate-speech datasets for Javanese either main unreleased [9] or conflate Javanese with broader Indonesian [10], [11]. Prior work has explored GPT-based data augmentation for low-resource, code-mixed Indonesian hate speech detection more broadly [12], [13], [14], including fine-tuning detectors on large samples of generated hate speech [15], most recently extending to multimodal Indonesian hate speech [16] and to explainability audits of detector behavior in Indonesian informatics journals [17], but without Javanese register-pragmatic structure as the organizing axis — the central contribution of this paper. In English, the difficulty of *implicit* hate for automated detectors is well documented [18], and machine generation

has been used to construct implicit-hate corpora at scale [19]; neither line of work, however, addresses register or diglossia, and no equivalent exists for Javanese.

The gap this paper addresses is therefore specific, and it persists in the most directly comparable recent work. A contemporaneous HateCheck-style functional test suite for Southeast Asian languages covers Indonesian, Tagalog, Thai, and Vietnamese but not Javanese [20], and the multilingual extension of HateCheck spans ten languages, none of them Indonesian regional languages [21]. Recent Indonesian hate-speech detection continues to target the national language, addressing low-resource conditions through semi-supervised augmentation of IndoBERT rather than through register structure [22]; work on the regional languages of other multilingual nations likewise organizes around language identity rather than variation *within* a language [23]. Across all of these, the axis of stratification is which language a text is in — never which register it is in. That is the axis this paper introduces.

A feature of Javanese that existing approaches entirely overlook is its *unggah-ungguh* (speech-level) system: the same hostile proposition can be expressed in coarse *ngoko* — the register of in-group bluntness — or in the refined *krama* register, producing qualitatively different pragmatic effects. Hot, direct rage is *ngoko*; cold contempt, ironic superiority, and indirect defamation — the forms that most damage group dignity — are *krama*. This is not a stylistic choice but a rule-governed pragmatic constraint, validated by a native speaker in this work.

The practical consequence is a data collection paradox. Social-media Javanese hate speech is almost entirely *ngoko* with code-mixing [24], [25]; pure *krama* is essentially absent from public posts because *krama* is used in formal or face-to-face contexts, not in Twitter or WhatsApp broadcasts [26]. A standard corpus-filtering pipeline will collect *ngoko* hate, confirm the register, and leave *krama* hate completely uncollected. We verified this directly: filtering 12,700 tweets from a public Indonesian hate-speech dump [27] yields 735 candidate Javanese texts, of which **zero** carry consensus *krama*-hate in the majority-vote labeling.

This scarcity is not an artifact of one corpus or one filtering method. The primary evidence for the collection paradox is the labeling run reported in Section 3.1, which is internal to this study and reproducible from its released artifacts; the measurement described next is corroborating evidence from an independent direction, and the argument does not rest on it. An unrelated companion measurement of Javanese digital presence — a lexicon-based scan of a general-topic (non-hate-filtered) Indonesian Twitter corpus spanning 32 cities and roughly 1.42 million cleaned, deduplicated tweets — confirms the same pattern from an independent angle [28]. Counting a tweet as confirmed-Javanese only if it contains two or more lexically distinctive Javanese function words or particles, just 1,321 tweets (0.093%) qualify. Table 1 places this in comparative context: among ten Indonesian regional languages surveyed in the same corpus, Javanese has both the largest speaker population (84.3 million, exceeding the other nine combined) and the *highest* confirmed-tweet rate — yet even this best case clears well under one tweet in a thousand. Two measurements that share no data, no method, and no base population — an LLM-based semantic filter over an already hate-labeled corpus (74 of 7,823 deduplicated tweets $\approx$ 0.9% [27]) and a lexicon-keyword filter over a general-topic corpus (1,321/1,419,641 $\approx$ 0.093% [28]) — converge on the same conclusion: Javanese is structurally underrepresented in exactly the public channels that hate-speech corpora are built from, regardless of how the filtering is done. This is not a contradiction of Javanese's 80-million-speaker population but consistent with it: sociolinguistic work on Javanese specifically argues that speaker-population size alone does not guarantee domain vitality, and documents a real, urbanization- and class-linked shift away from Javanese (particularly its *krama* register) toward Indonesian in exactly the informal, everyday domains that social media represents [29], [30].

Table 1. Confirmed-tweet rate for ten Indonesian regional languages in a general-topic, 32-city Twitter corpus ($\approx 1.42\text{M}$ cleaned tweets; lexicon match, ≥ 2 distinctive-word threshold per tweet) [28]. Not to be confused with the hate-detector verdict rates in Table 8.

Language	Speakers (M)	Confirmed tweets	Confirmed rate
Javanese	84.3	1,321	0.093%
Acehnese	3.5	515	0.036%
Banjarese	3.5	302	0.021%
Minangkabau	5.5	286	0.020%
Sundanese	34.0	256	0.018%
Balinese	3.3	81	0.006%
Ngaju	0.9	71	0.005%
Toba Batak	2.0	38	0.003%
Madurese	7.0	11	0.0008%
Buginese	5.0	2	0.0001%

This paper addresses the collection paradox with a diagnostic approach analogous to functional test suites for hate speech detection [31]: rather than filtering and labeling an existing corpus, we construct a register-stratified set of Javanese hate-speech stimuli [32] — using large language models (LLMs) as the construction *method*, not as the object of study — validate each stimulus with a native expert, and probe the set with five hate-speech detector models. The contribution is not a large dataset but a controlled diagnostic instrument: a minimal sufficient set that proves a specific failure mode exists, is reproducible, and is systematically missed by current automated detection. Generation is the *means* by which the otherwise-uncollectable *krama* niches become testable; the finding that matters is what the resulting stimuli reveal about detection, not the stimuli themselves.

The objective of this study is therefore to diagnose and characterize the register-pragmatic blind spot in Javanese hate speech detection: to establish that *krama*-borne hostility is structurally uncollectable from social media, to construct and natively validate a register-stratified stimulus set that makes it testable, and to measure how far current detector models fail on it. This objective yields four contributions.

Contributions. (1) Empirical proof of a *detection blind spot*: Javanese ironic hate (*pasemon* in *krama*) evades all five tested detectors — cloud and local — at a rate of 96%, while explicit *ngoko* hate is caught at 100%; the failure traces to pragmatic implicature, not surface politeness. (2) A *register-pragmatic model* of Javanese hate speech — a taxonomy of four niches defined by the interaction of register and pragmatic mode, validated with a native expert — that explains *why* the blind spot exists, not merely that it does. (3) A register-stratified diagnostic stimulus set, constructed via LLM generation and validated for native authenticity, that makes the otherwise-uncollectable *krama* niches testable for the first time; generation here is a construction method in service of diagnosis, not an end in itself, and the resulting set is released under a restricted, research-only license precisely because it demonstrates how current detectors can be evaded. (4) A publicly released codebook, diagnostic construction scripts, and evaluation pipeline for reproducible extension to other diglossic, register-rich languages.

2. METHOD

The research proceeds in five stages, shown in Figure 1: corpus filtering, a labeling baseline that establishes the scarcity of *krama* hate, construction of the diagnostic stimulus set, native authenticity validation, and the five-detector blind-spot probe. The first two stages establish the collection paradox that motivates the design; the last three construct the instrument and measure the blind spot. Each stage is described in turn in Sections 2.1 to 2.8.

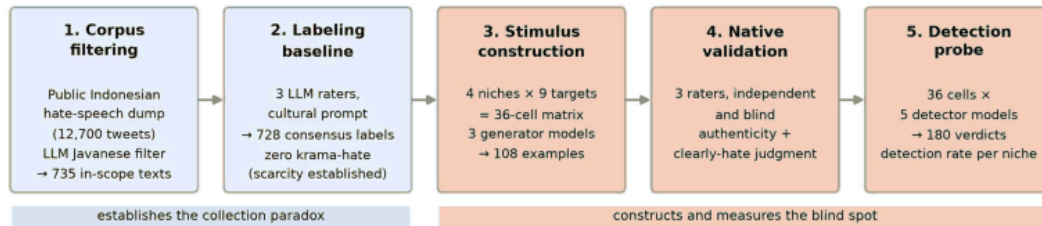


Figure 1. Research pipeline. Stages 1–2 establish the collection paradox by measuring how little *krama* hate is recoverable from public text; stages 3–5 construct the register-stratified stimulus set, validate its authenticity with native raters, and probe five detector models. Counts shown are the actual yields reported in Sections 3.1 to 3.5.

11 2.1. Hate speech definition

We define **hate speech** operationally as **text** that **attacks, demeans, dehumanizes, or incites violence or discrimination against a person or group based on group identity** — ethnicity (*suku*), **religion** or belief (*agama/kepercayaan*), gender or sexual orientation, social class, or political collective. The decisive criterion is *direction of attack toward group identity*, not coarseness of language. Javanese is rich in profanity (*pisuhan*: *asu, jancuk, cok*) that functions as in-group bonding in the Arek/Surabaya register; coarseness alone is not hate. This distinction is the single most error-prone boundary in Javanese moderation.

2.2. Four-dimensional taxonomy

The taxonomy has four dimensions (full definitions, decision trees, and worked examples in the released codebook):

Target group — ethnicity (*suku* *Madura, Tionghoa, Sunda, Batak, Arab, ...*), intra-Javanese region (*Mataraman, Arek, Banyumasan*), religion or belief, gender/LGBTQ+, political collective, or *tidak ada*.

Severity — *BUK* (not hate) / *ringan* (light: stereotyping) / *sedang* (moderate: dehumanization, exclusion) / *berat* (severe: threats, incitement).

Register — *ngoko* / *madya* / *krama* / *campur kasar*. This dimension is the paper's primary focus; see Section 2.3.

Form — *direct* / *sarcastic* / *idiomatic* (*pasemon*) / *code-switched*.

2.3. The register-pragmatic model: four niches

Hostility in Javanese is not register-flat. The interaction between speech level and pragmatic intent produces four empirically distinct niches (Table 2, Figure 2), validated by a native Javanese speaker:

Table 2. The four register-pragmatic niches: register, pragmatic mode, and mechanism.

Niche	Register	Pragmatic mode	Mechanism
N1 <i>ngoko direct</i>	ngoko/kasar	Hot, open aggression	Explicit slur + profanity to addressee or about group
N2 <i>krama report</i>	krama alus	Derogatory report, absent target	Polite prayer/concern framing derogate-third-party ("Mugi tiyang X enggal...")
N3a <i>krama sarcastic</i>	krama alus	Ironic over-praise (<i>pasemon</i>)	Mock-deference: weaponized honorifics ("Panjenengan wicaksana sanget...")

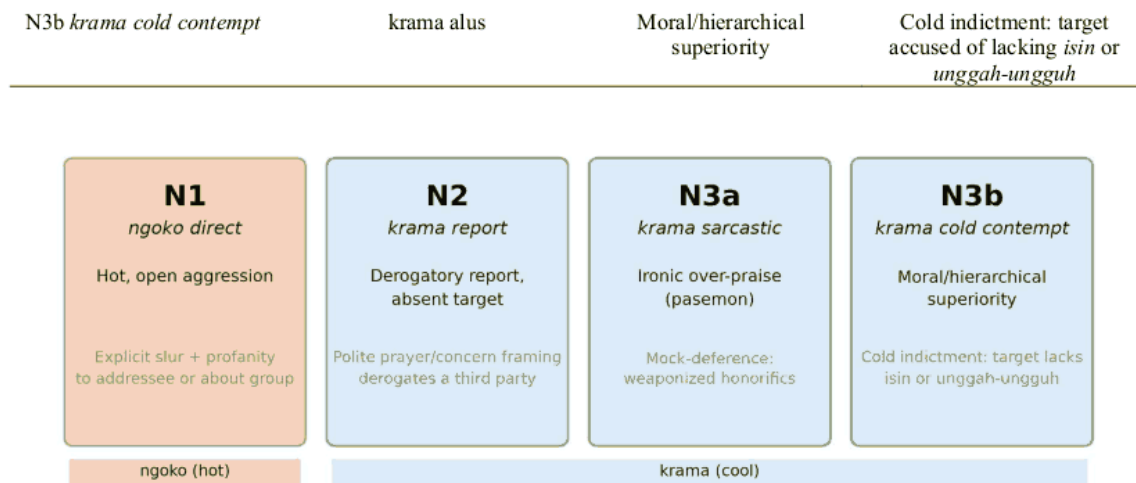


Figure 2. The four register-pragmatic niches (mechanism detail as in Table 2), grouped by register: *ngoko* (hot, direct) vs. *krama* (cool, indirect), the latter spanning three distinct pragmatic modes.

A key empirical finding (Section 3.1) is that N2–N3b are essentially absent from social-media data — confirmed by zero consensus *krama*-hate in 728 labeled real texts — establishing the collection paradox and motivating generation.

2.4. Why *krama* hate is uncollectable

Two structural factors make *krama* hate unrepresented in social-media corpora. First, *diglossia*: *krama* is the register of formal, elder-addressee, or face-to-face speech; social-media Javanese defaults to *ngoko* with Indonesian code-mixing [26]. Second, *irony recognition* [33], [34]: even if a *krama-sarcastic* post were found, an automated classifier would need to resolve pragmatic implicature (*pasemon*) — a specifically Javanese indirect speech act — rather than surface lexical hostility. Section 3.5 shows this is beyond all five tested detectors.

2.5. Evidence of scarcity: the labeling baseline

Before generation, we establish the scarcity empirically. We filter a public Indonesian hate-speech dump [27] for Javanese and code-mixed text using an LLM filter (validated in a 250-tweet pilot: 100% JSON validity, 9.6% Javanese yield), apply a two-stage cascade (free local pre-screen → cloud verification), and label the resulting pool with three LLM raters (DeepSeek, Grok, Qwen3-14B/local) using a culturally-grounded prompt [35], [36].

The filter itself is a single-model classifier (Grok, prompt `jawa\_filter\_v0`) that assigns each tweet one of four language categories — *jawa* (majority-Javanese lexicon and structure), *campuran* (a genuine Javanese phrase or clause code-mixed with Indonesian), *indonesia*, or *lainnya* — along with a confidence score, a register label, and the list of Javanese markers it found. Three design decisions matter for replication. First, the prompt enumerates Javanese function words and particles by register (*kowe, ora, arep, opo, piye, wis, durung* for *ngoko*; *panjenengan, kula, badhe, mboten, sampun, dereng* for *krama*) so that the decision rests on named morphological evidence rather than on an opaque language-identification score. Second, it explicitly instructs the model that widespread Indonesian slang loanwords (*banget, aku, apa, gak, nih, sih, dong*) are not on their own evidence of Javanese — the dominant false-positive source when off-the-shelf language identification is applied to Indonesian social media. Third, Javanese-specific expletives are listed as strong positive evidence, because in a hate-speech dump the clearest Javanese signal is frequently carried by profanity. The inclusion criterion for

the working pool is category *jawa* OR *campuran*; both are retained because code-mixed text is the empirical norm for Javanese social media rather than a degenerate case.

Applied to the full dump, the filter returns parseable output for 6,040 of 7,823 deduplicated tweets and classifies 735 of those (12.2%) as in-scope — 74 majority-Javanese (1.2%) and 661 code-mixed (10.9%). We note the 1,783 unparseable responses (22.8%) explicitly, since the 250-tweet pilot had returned 100% JSON validity: strict-JSON reliability measured on a small pilot did not survive scale-up, which is itself a practical finding for LLM-filtering pipelines. Unparseable responses were discarded rather than retried, so the reported yield is a lower bound on the Javanese content of the dump. Inter-rater reliability throughout this paper is measured with Krippendorff's alpha [37], defined in (1):

$$\alpha = 1 - \frac{D_o}{D_e} \quad (1)$$

where D_o is the observed disagreement among raters and D_e is the disagreement expected by chance; alpha = 1 indicates perfect agreement and alpha = 0 chance-level agreement. The cascade's cloud confirmation rate of **23.9%** (403 of 1,687 pre-screen survivors) is itself a finding: cheap local models reduce a candidate pool but cannot replace cloud precision for filtering.

The resulting pool yields **728 consensus-labeled instances** (158 hate, 21.7%; α Krippendorff [37] 0.51 three-rater held-out, 0.69 for the cloud pair), of which **register is ngoko for 157/158 hate instances** and *krama*-hate consensus is zero. This negative result serves two purposes: it motivates generation by proving the collection paradox, and the 728 texts serve as a realism anchor for generator calibration.

Full methodology of the labeling pipeline — prompt engineering ($\Delta\alpha \approx +0.23$ from two definition corrections), vendor selection, cascade economics, and adversarial verification — is documented in the supplementary report. The core finding relevant to this paper: **social-media filtering cannot produce *krama* hate at meaningful scale, even from a 12,700-tweet dump.**

2.6. Constructing the diagnostic stimulus set

Why generation rather than annotation. The design of this stage is directly motivated by a prior failure. In a precursor project, student annotators produced nominally "Javanese" data primarily by back-translating Indonesian and English hate speech, yielding labels that were internally consistent but not grounded in authentic Javanese expression; the corruption was invisible until downstream models failed to generalize. Constructing stimuli and validating them removes that failure mode by design: the human role collapses to *authenticity refereeing* — a native speaker judges whether generated text sounds like real Javanese hate — which requires native comprehension rather than sustained annotation labor, and is therefore far less vulnerable to the shortcut that corrupted the earlier effort.

We construct a **4-niche × 9-target = 36-cell matrix** of Javanese hate-speech stimuli using DeepSeek [38] as the primary construction model and Gemma3-27B [39] and Qwen3-14B [40] (both running locally) for comparison, producing **108 total examples** (3 per cell). The nine target groups span the Indonesian SARA taxonomy (*suku*, *agama*, *ras*, *antargolongan* — ethnicity, religion, race, and inter-group relations): *suku* (Madura, Tionghoa, Arab), *agama* (Islam, Kristen), *gender* (wanita, LGBTQ+), *politik* (collective), and intra-Javanese (*Arek vs. Mataraman*).

Each generation call specifies: (a) niche (N1–N3b), (b) target group, (c) a culturally-grounded system prompt (in Indonesian) that defines the niche's pragmatic mechanism and provides two-to-four few-shot examples. The system prompt explicitly prohibits "museum *krama*" (literary-Javanese vocabulary unused in living speech) and Indonesian code-leak. Generation uses the same API infrastructure as the labeling pipeline. The specific API endpoint used ('deepseek-v4-pro') postdates the cited DeepSeek-V3 technical report [38]; no separate V4 technical report could be independently verified at the time of writing, so [38] is cited as the closest documented architecture family rather than an exact version match.

Inference parameters. Table 3 lists the decoding settings used at each stage, as executed. Generation deliberately uses a non-zero sampling temperature (0.8), since the task requires lexical and idiomatic variation across cells; every other stage — labeling and detection — uses greedy decoding (temperature 0) so that verdicts are as close to deterministic as the vendor APIs permit. Generation calls escalate the output-token budget on truncation (12,000 → 20,000 → 30,000 tokens, retrying when the response is empty or returns `'finish_reason = "length"'`). This escalation is not incidental: `'deepseek-v4-pro'` is a reasoning model that can consume an entire modest budget on internal reasoning tokens and return empty visible content, which is what a fixed smaller budget produced in pre-run trials. The reasoning-capable local model (Qwen3-14B) is invoked with a `'/no_think'` directive to suppress the same behaviour. We note as a reproducibility limitation that none of the vendor endpoints used exposes a random-seed parameter, so generation is reproducible in distribution rather than bit-exactly; the labeling and detection stages, being greedy, are substantially more stable across runs. All released scripts carry these settings as defaults.

Table 3. Inference parameters by pipeline stage, as executed.

Stage	Model(s)	Temperature	Max output tokens	Prompt version
Javanese language filter	Grok	0.0	1,024	<code>'jawa_filter_v0'</code>
Labeling (3 raters)	DeepSeek / Grok / Qwen3-14B	0.0	2,048 / 1,024 / 4,096	<code>'cultural_classification_v2'</code>
Stimulus generation (cloud)	DeepSeek	0.8	12,000 → 20,000 → 30,000 (escalating)	generation system prompt, v2
Stimulus generation (local)	Gemma3-27B / Qwen3-14B	0.0	4,096	generation system prompt, v2
Detection probe (5 detectors)	DeepSeek / Grok / Qwen3-14B / Gemma3-27B / GPT-OSS-20B	0.0	2,048 / 1,024 / 4,096 / 4,096 / 4,096	<code>'cultural_classification_v2'</code>

2.7. Native authenticity validation

The 108 generated examples are submitted to the first author (native Javanese speaker from East Java) for authenticity validation. The validation instrument is a structured spreadsheet with: (a) the generated text, (b) the intended niche/target, (c) the generation mechanism description, (d) a machine-caught flag (did the production detector identify it as hate?), (e) an auto-concern flag (QC panel pre-triage), and (f) a binary authenticity verdict (1 = authentic Javanese hate of the stated type, 0 = not authentic). Comments on specific issues are recorded for each rejected example.

The human role is explicitly *authenticity refereeing*: does this text sound like something a Javanese speaker would actually say in the stated register-pragmatic mode? This is a comprehension task, not a production or annotation task, and can be completed in one to two hours across all 108 examples.

A second and third rater subsequently repeated the full 108-item validation independently and blind: co-author Yekti Asmoro Kanthi (native Javanese speaker) and co-author Daniel Rudiaman Sijabat (a 30-year East Java resident, highly proficient in Javanese as an additional language, not a native speaker). Their instrument separates the original single judgment into two binary columns: OTENTIK ("does this sound like authentic Javanese in the stated register?") and JELAS_HATE ("independently of authenticity, does this clearly read as an attack on a group identity?"). Neither rater saw the first author's verdicts or each other's answers before both were complete. This two-column design was introduced after the first author's single-column pass; the consequences of that instrument difference are analyzed in Section 4.3.

2.8. Detection blind-spot probe

We run each of the 36 DeepSeek-generated cells through five detector models: the production three-rater set (DeepSeek, Grok, Qwen3-14B) plus two additional local models (Gemma3-27B [39], GPT-OSS-20B [41]). Each detector is given the same prompt v2 (the production prompt from Section 2.5) and asked to return a binary hate/non-hate verdict — the zero-shot prompted-LLM detection setup now standard in hate speech research [42]. The detection rate of detector d on niche n is defined in (2):

$$DR_{n,d} = \frac{1}{|C_n|} \sum_{c \in C_n} 1[v_d(c) = 1] \quad (2)$$

where C_n is the set of nine stimulus cells in niche n and $v_d(c) \in \{0, 1\}$ is detector d 's binary hate verdict on cell c . Detection rate per niche is the fraction of cells where the detector returns hate=True. Every detection rate reported in Section 3.5 is accompanied by a Wilson score interval, chosen over the normal approximation because several cells sit at 0% or 100% where the normal approximation is degenerate. Contrasts between niches are reported as differences in proportions with percentile bootstrap 95% confidence intervals over 10,000 resamples at a fixed seed; the same procedure is applied to the between-generator authenticity contrasts in Section 3.2. We report intervals rather than null-hypothesis significance tests throughout, because the quantity of interest is the size of the detection gap rather than its distinguishability from zero, and because the 36-cell design was fixed in advance for coverage of the register $\times$ target matrix rather than powered for a hypothesis test. We use "five detectors" throughout to mean five distinct model checkpoints, not five architecturally or taxonomically independent detection systems — all five share prompt v2's hate definition, so the probe measures whether different models converge or diverge under one shared taxonomy, not whether the taxonomy itself generalizes to detectors trained or prompted differently (Section 4.6, Limitation 5). None of the five is a fine-tuned classifier or a deployed commercial moderation system.

The probe is controlled: the generator is held constant (DeepSeek), so differences in detection rate across niches reflect niche difficulty, not generator variation. All 180 (cell $\times$ detector) verdict calls succeeded with no parse failures (verified by independent recount).

3. RESULT

3.1. Supporting results: labeling baseline confirms scarcity

The 728 labeled social-media texts (Section 2.5) provide the empirical grounding for the generation argument. Register is *ngoko* for 157 of the 158 hate instances and coarse code-mixed (*campur kasar*) for the remaining one; no consensus-hate instance is *krama*. The corpus's only two *krama*-register texts are non-hate, with no group target. *Krama*-hate consensus is zero. This is not a trivial finding — it directly rules out the alternative explanation that *krama* hate is collectable in volume and that generation is unnecessary. The labeling pipeline achieves a moderate cross-model agreement (cloud pair α 0.69 held-out, three-rater α 0.51 held-out), demonstrating that the multi-LLM consensus approach is reliable for the types of hate it can collect, while leaving the *krama* register uncovered by design of the data source.

The comparison with the source dataset's human labels (agreement 54.5%, $\kappa = 0.19$) reflects a definitional difference rather than an error [43]: the cultural prompt deliberately narrows hate to group-directed attacks, excluding bare profanity, and flags genuine Javanese identity slurs missed by an Indonesian-context annotator. This alignment between the labeling baseline and the generation taxonomy is methodological consistency, not an artifact.

3.2. Generation authenticity: overall and by model

Of 108 generated examples, 59 (55%) were judged authentic Javanese hate of the stated type. Table 4 breaks this down by generator.

Table 4. Native authenticity rate by generator model.

Generator	Authentic	Rate	Primary failure mode
DeepSeek	35/36	97%	1 item: <i>krama-sarcastic</i> read as sincere blessing, not irony
Gemma3-27B (local)	20/36	56%	<i>Krama-sarcastic</i> : Indonesian leak in sarcastic punchline (0/9)
Qwen3-14B (local)	4/36	11%	<i>Krama</i> registers: generates Indonesian (not Javanese); hallucinated word <i>kacandran</i>

The ordering is not an artifact of small samples. All three pairwise contrasts exclude zero: DeepSeek exceeds Gemma3-27B by 41.7 percentage points (95% bootstrap CI [25.0, 58.3]), DeepSeek exceeds Qwen3-14B by 86.1 pp (CI [75.0, 97.2]), and Gemma3-27B exceeds Qwen3-14B by 44.4 pp (CI [25.0, 63.9]). The per-generator rates carry Wilson 95% intervals of [85.8%, 99.5%] for DeepSeek, [39.6%, 70.5%] for Gemma3-27B, and [4.4%, 25.3%] for Qwen3-14B — wide, as 36 items per generator implies, but non-overlapping between the best and worst models.

DeepSeek is the only generator that can produce authentic Javanese in all four niches. The single failure (example #21, *krama-sarcastic* targeting *agama_islam*) is itself informative: an ironic over-praise of a santri's nighttime prayers was judged by the native speaker as a *sincere* blessing — the sarcasm failed entirely, producing a false positive in the validator's favor. This documents the difficulty of the N3a niche even for the best-performing generator.

This ranking is robust to which native validator scores it: splitting Table 4 by validator (Mukhlis / Yekti / Daniel), DeepSeek ranks first for all three (97% / 100% / 97%), and the full ordering DeepSeek > Gemma3-27B > Qwen3-14B holds for all three despite Yekti's generally more lenient absolute rates (97% / 56% / 11% vs. Yekti's 100% / 97% / 75% and Daniel's 97% / 39% / 0%; full breakdown in the supplementary materials). The headline generator-quality claim survives multi-rater scrutiny even though the pooled 55% authenticity figure does not (Section 3.4).

3.3. Register difficulty: *krama-sarcastic* is the hardest niche

Table 5 shows authenticity by niche, pooled across generators.

Table 5. Native authenticity rate by register-pragmatic niche (all generators combined, n=27 per niche).

Niche	Authentic	Rate
N1 <i>ngoko direct</i>	22/27	81%
N2 <i>krama report</i>	17/27	63%
N3a <i>krama sarcastic</i>	8/27	30%
N3b <i>krama cold contempt</i>	12/27	44%

N1 (*ngoko direct*) is easiest — explicit profanity and slur targeting are within all generators' capability. N3a (*krama sarcastic*) is hardest at 30%; the Gemma3 failures account for nearly all the deficit (Gemma3 produced 0/9 authentic *krama-sarcastic* examples due to Indonesian leak). DeepSeek achieves 8/9 (89%) on this hardest niche; the single failure is example #21 above.

The N3b (*krama cold contempt*) result is particularly meaningful: this niche — a group-directed moral indictment in refined *krama* — was an open research question at the start of this work (could LLMs generate authentic *krama* hate directed at a SARA group, not just at individuals?). The answer is **yes, at 44% overall and 9/9 (100%) for DeepSeek**. The DeepSeek N3b examples use a recurring device — accusing the target group of lacking *isin* (shame/modesty) or *unggah-ungguh* (proper

etiquette) — which the native speaker judged authentic and recognizable as a real Javanese contempt pattern. The formulaic character of this device is itself a finding, suggesting LLMs learned it from literary and religious Javanese text.

3.4. Multi-validator authenticity

The authenticity rates reported in Section 3.2–3.2 (Table 4, Table 5) come from the first author (native Javanese speaker). A second and third independent, blind pass on the same 108 items was subsequently completed by co-author Yekti Asmoro Kanthi (native Javanese speaker) and co-author Daniel Rudiaman Sijabat (a long-term East Java resident of 30 years, highly proficient in Javanese as an additional language, but not a native speaker). Each judged independently, blind to the first author's verdicts and to each other's answers. Table 6 reports the result.

Table 6. Per-validator authenticity rate and pairwise Krippendorff's α (raw OTENTIK judgment).

	Authentic rate	vs. Mukhlis	vs. Yekti	vs. Daniel
Mukhlis	55% (59/108)	—	0.095	0.779
Yekti	91% (98/108)	0.095	—	-0.039
Daniel	45% (49/108)	0.779	-0.039	—

Computed with (1), 3-rater $\alpha = 0.336$ (95% CI [0.224, 0.449]) — below the conventional 0.667 threshold [37].

This does not confirm the 55% figure as a stable consensus. Notably, the two participants who are both native Javanese speakers (Mukhlis, Yekti) disagree at a level statistically indistinguishable from chance, while Daniel — despite not being a native speaker — tracks Mukhlis's judgments closely. We read this as a finding rather than a nuisance result: authenticity, as operationalized by a binary refereeing task, is not a high-consensus construct even between native Javanese speakers, and the 55%/97%/11% generator rates in Table 4 should be read as one qualified evaluator's estimate rather than an inter-subjectively validated ground truth. We therefore report all three rates and the full pairwise- α matrix rather than pooling them, and we do not use Daniel's agreement with Mukhlis to claim generalizability, since his linguistic background differs from the two native raters'.

Table 7 tests whether this is an instrument artifact by recomputing agreement with Yekti's and Daniel's two answers combined (authentic AND clearly-hate — approximating the single question Mukhlis's original one-column instrument had to answer at once).

Table 7. Raw vs. harmonized (OTENTIK AND JELAS_HATE) Krippendorff's α .

Pair	Raw α	Harmonized α
Mukhlis–Yekti	0.095	0.519
Mukhlis–Daniel	0.779	0.448
Yekti–Daniel	-0.039	0.609

Figure 3 visualizes the per-validator rates by generator: the model ordering is identical for all three validators (compare Section 3.2).

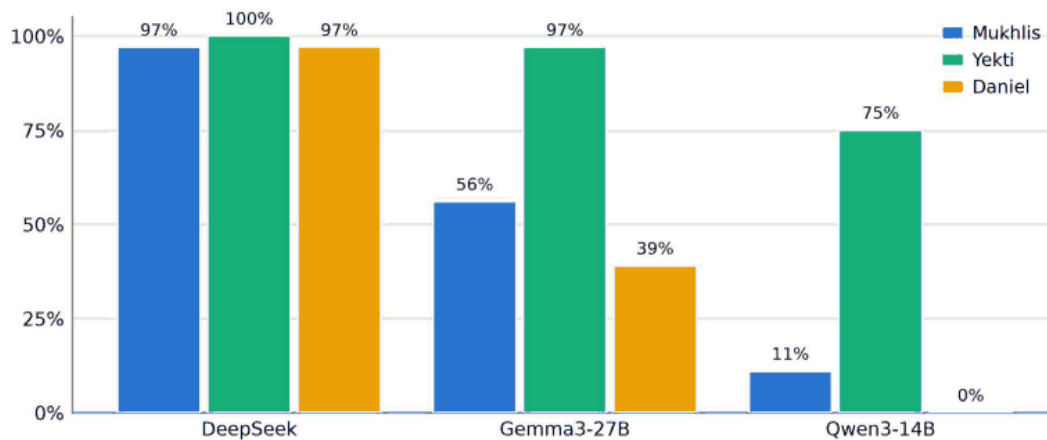


Figure 3. Authenticity rate by generator and validator. The DeepSeek > Gemma3-27B > Qwen3-14B ordering holds for all three validators despite different absolute leniency.

3.5. Detection blind spot: pasemon evades every detector

Table 8 shows detection rates across all five detectors for each niche (Figure 4), computed with (2).

Table 8. Hate detection rate by niche and detector (DeepSeek-generated cells; n=9 cells per niche per detector).

Niche	DeepSeek	Grok	Qwen3-14B	Gemma3-27B	GPT-OSS-20B
N1 <i>ngoko direct</i>	100%	100%	100%	100%	100%
N2 <i>krama report</i>	78%	89%	44%	89%	44%
N3a <i>krama sarcastic</i>	11%	11%	0%	0%	0%
N3b <i>krama cold contempt</i>	78%	89%	56%	78%	78%

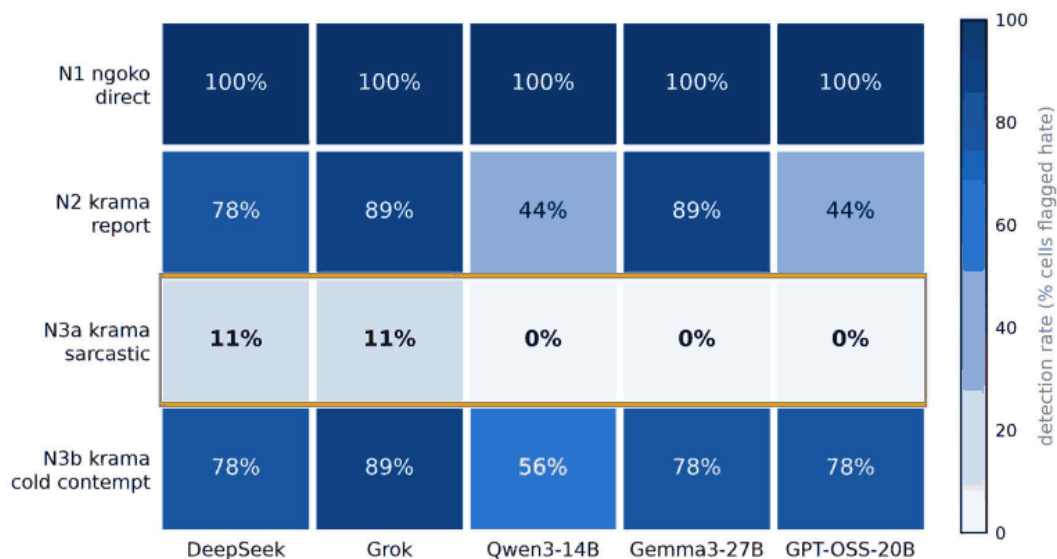


Figure 4. Table 8 as a heatmap. The *krama-sarcastic* row (outlined) is visibly near-white across all five detector models, while every other niche is caught at 44–100%.

Figure 5 presents the same detection rates as grouped bars with Wilson 95% intervals, so that the uncertainty attaching to each nine-cell estimate is visible alongside the point estimate. The intervals are wide, as a nine-cell denominator implies, but the *krama-sarcastic* cluster is separated from every other niche by a margin far exceeding that width for all five detectors.

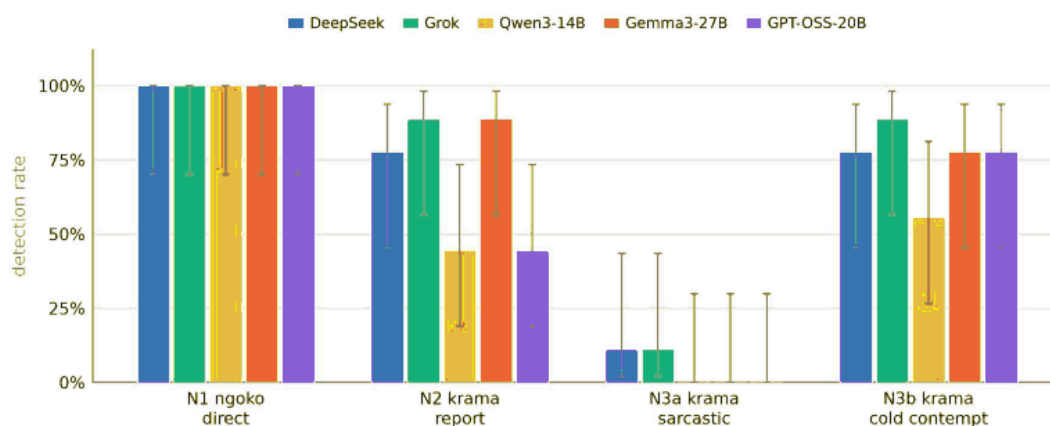


Figure 5. Detection rate by niche and detector model with Wilson 95% confidence intervals ($n = 9$ cells per bar). Bars at zero carry an interval but no visible bar. The N3a *krama-sarcastic* cluster sits at or near zero for every detector, while the two other *krama* niches remain comparable to *ngoko-direct* detection — the contrast that distinguishes pragmatic implicature from surface politeness as the source of the failure.

The N3a result is striking: across 45 total detector verdicts on *krama-sarcastic* cells, only 2 (4%) correctly identify the content as hate (Wilson 95% CI [1.2%, 14.8%]). This holds for both cloud models (11% each) and all three local models (0%). Because each per-niche rate rests on 45 verdicts, the percentages in this section should be read as indicative of the direction and rough magnitude of an effect rather than as precise point estimates, and we report intervals alongside them throughout. The contrast that carries this paper's central claim is nonetheless large enough to survive the sample size: the gap between *ngoko-direct* detection (45/45, Wilson 95% CI [92.1%, 100%]) and *krama-sarcastic* detection is 95.6 percentage points (95% bootstrap CI [88.9, 100.0]), and the gaps against *krama-report* (64.4 pp, CI [48.9, 77.8]) and *krama-cold-contempt* (71.1 pp, CI [57.8, 84.4]) likewise exclude zero. All intervals are percentile bootstrap estimates over 10,000 resamples with a fixed seed, computed by a released script. N1 (*ngoko direct*) is caught at 100% everywhere — explicit hostility is trivial. The detection challenge is not *surface politeness* (N2 *krama-report* is caught at 78–89% by cloud models) but *pragmatic implicature*: an ironic over-compliment in *krama* does not contain lexically hateful tokens, only the pragmatic inversion of apparent praise into attack.

Nine of the 36 cells were missed by all five detectors simultaneously: seven of the nine *krama-sarcastic* cells, plus one *krama-report* and one *krama-cold-contempt* cell, both targeting *politik_kolektif* (polite political derogation reads as ordinary criticism to all five raters). These 9 cells represent the extremity of the blind spot — hate that the entire automated detection ecosystem cannot see. Section 4.2 revisits these nine cells individually, using the validators' independent JELAS_HATE judgments, and narrows the blind-spot claim to the five that target a SARA identity axis; the other four evade detection for reasons we disclose separately (Section 4.6, Limitation 6). The two counts are consistent rather than contradictory — nine cells evade every detector, of which five support the pragmatic-blindness reading — and we flag the distinction here so it is visible to a reader of the Results alone.

A notable caveat: DeepSeek, which generated the *krama-sarcastic* attacks to an explicit hate specification, then detects only 11% of them when serving as a rater. This shows that generation